

WiFi Repeater Placement in Urban Networks

Exact MIP Formulation (Gurobi) · Simulated Annealing Metaheuristic · Comparative Analysis

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1. Problem Statement

Given a city modelled as an $n \times n$ integer grid, a set of clients J requiring wireless coverage, and a set of candidate repeater locations V with associated placement costs, we seek the **minimum-cost** subset of locations at which to install repeaters such that:

- every client is within Euclidean radius R of at least one active repeater (*coverage*);
- all active repeaters form a single connected component rooted at a fixed source node v_0 (*backbone connectivity*).

The problem is a variant of the **Capacitated Facility Location Problem (CFLP)** augmented with a graph-connectivity constraint, and is NP-hard by reduction from SET COVER.

2. Mathematical Formulation

2.1. Sets & Parameters

Symbol	Meaning
$V = \{0, \dots, n^2 - 1\}$	Candidate locations; $v_0 = 0$ is source
$J = \{1, \dots, m\}$	Client set
$E \subseteq V \times V$	Undirected edges: $(i, k) \in E$ if $d_{ik} \leq \max(r_i, r_k)$
A	Directed arcs: both (i, k) and (k, i) for each $(i, k) \in E$
$c_i \in \mathbb{R}^+$	Placement cost at location i
$R, R_S \in \mathbb{R}^+$	Coverage radius (repeater / source)
r_i	R_S if $i = 0$, else R
$\text{cov}(i) \subseteq J$	Clients within r_i of location i
M	Big-M constant = $ V - 1$

2.2. Decision Variables

Var	Domain	Meaning
x_i	$\{0, 1\}$	1 if repeater placed at i
y_{ij}	$\{0, 1\}$	1 if client j served by repeater i
z_{ik}	$\{0, 1\}$	1 if undirected edge $(i, k) \in E$ active
f_{ik}	$\mathbb{R}_{\geq 0}$	Flow on directed arc $(i, k) \in A$
s_0	$[0, M]$	Slack supply injected at source v_0

2.3. Objective Function

$$\min_x \sum_{i \in V} c_i \cdot x_i \quad (\text{P})$$

Minimise total deployment cost. Post-solve profitability is reported as $\pi_i = (\sum_j y_{ij}) / c_i$.

2.4. Constraints

(C0) Source always active

$$x_0 = 1 \quad (\text{C0})$$

(C1) Full client coverage

$$\sum_{i \in V} y_{ij} \geq 1 \quad \forall j \in J \quad (\text{C1})$$

(C2) Assignment to active repeaters only

$$y_{ij} \leq x_i \quad \forall i \in V, j \in \text{cov}(i) \quad (\text{C2})$$

(C3) Edge activation (tightened)

$$z_{ik} \leq x_i, \quad z_{ik} \leq x_k, \quad z_{ik} \geq x_i + x_k - 1 \quad \forall (i, k) \in E \quad (\text{C3})$$

The third inequality forces $z_{ik} = 1$ whenever both endpoints are active, tightening the LP relaxation.

(C4) Flow only on active arcs

$$f_{ik} \leq M \cdot z_{e(i,k)} \quad \forall (i, k) \in A \quad (\text{C4})$$

where $e(i, k)$ maps each directed arc to its undirected edge.

(C5) Single-commodity flow (connectivity)

$$s_0 + \sum_{k:(k,0) \in A} f_{k0} - \sum_{k:(0,k) \in A} f_{0k} = x_0 \quad (\text{C5a})$$

$$\sum_{k:(k,i) \in A} f_{ki} - \sum_{k:(i,k) \in A} f_{ik} = x_i \quad \forall i \in V \setminus \{0\} \quad (\text{C5b})$$

The source v_0 injects one unit of flow per active node; every active repeater i absorbs exactly one unit. An active repeater can receive flow only through an active edge, so any disconnected component cannot balance its flow, making the constraint infeasible unless all active nodes are reachable from v_0 .

(C6) Domain: $x_i, y_{ij}, z_{ik} \in \{0, 1\}$; $f_{ik} \geq 0$; $s_0 \in [0, M]$

3. Gurobi — Exact MIP Solution

3.1. Branch-and-Bound

Gurobi solves (P) via **Branch-and-Bound** with LP relaxation. Binary variables are relaxed to $[0, 1]$ for a lower bound; fractional variables are branched ($x_i=0$ / $x_i=1$); nodes whose LP bound exceeds the incumbent are pruned. Cover, MIR, flow-cover, RLT, and network cuts tighten the relaxation. Termination is at MIP gap $\leq 10^{-4}$ (proven optimal). The instance yields **803 variables** and **1246 constraints** ($n=8$, $|J|=15$); Gurobi explored 983 B&B nodes and applied 249 cutting planes.

3.2. Key Implementation

```
m = Model("wifi_repeater")
x = m.addVars(num_locs, vtype=GRB.BINARY) # C0-C2
y = m.addVars([(i,j) for i in V for j in cov[i]], vtype=GRB.BINARY)
z = m.addVars(edges, vtype=GRB.BINARY) # C3
f = m.addVars(directed, vtype=GRB.CONTINUOUS, lb=0) # C4-C5
s0 = m.addVar(vtype=GRB.CONTINUOUS, lb=0, ub=BM)

m.setObjective(quicksum(costs[i]*x[i] for i in V), GRB.MINIMIZE)
m.addConstr(x[0] == 1) # C0
for (i,k) in edges:
    m.addConstr(z[i,k] >= x[i] + x[k] - 1) # C3 (tightened)
# C5: SCF flow balance one unit per active node
m.addConstr(s0 + quicksum(f[k,0] for k in in_arcs[0])
             - quicksum(f[0,k] for k in out_arcs[0]) == x[0])
for i in V:
    if i==0: continue
    m.addConstr(quicksum(f[k,i] for k in in_arcs[i])
               - quicksum(f[i,k] for k in out_arcs[i]) == x[i])
m.optimize()
```

4. Metaheuristic — Simulated Annealing

4.1. Encoding & Neighbourhood

The solution is a binary vector $\mathbf{x} \in \{0, 1\}^{|V|}$ with $x_0 = 1$ fixed. Each neighbour is generated by flipping one randomly chosen bit x_i ($i \neq 0$), followed by a two-phase **repair**:

1. **Coverage repair.** For every uncovered client j , activate the cheapest node in $\text{rev_cov}(j)$.
2. **Connectivity repair.** Run BFS from v_0 over active nodes. While orphan (active but unreachable) nodes remain, activate the cheapest inactive neighbour of any reachable node, growing the reachable component until all active nodes are connected.

Repair guarantees feasibility after every flip, so the best recorded solution is always valid.

4.2. Acceptance Criterion & Parameters

A move is accepted with probability $\min(1, e^{-\Delta/T})$, where $\Delta = f_{\text{new}} - f_{\text{curr}}$ and f is the *penalised cost*:

$$f(\mathbf{x}) = \sum_i c_i x_i + 10^3 \cdot \#(\text{uncovered clients}) + 10^3 \cdot \#(\text{disconnected active nodes})$$

Parameter	Value
Initial temperature T_0	50.0
Minimum temperature $T_{\min}$	0.01
Cooling rate α	0.995
Max iterations	30 000

4.3. Complexity

	Gurobi	SA
Time (worst)	$O(2^{ V })$	$O(T \cdot V \cdot J)$
Memory	$O(V + J + E)$	$O(V + J + E)$
Optimality	Proven	Approx.

5. Results

The instance uses $n=8$ (64 candidate locations), 15 clients, $R=1.5$, $R_S=2.5$, random seed 43. Gurobi explored 983 B&B nodes and applied 249 cutting planes (cover, MIR, flow cover, RLT, network) before certifying optimality.

Metric	Gurobi	SA
Total cost	36.89	42.10
Repeaters placed	13/64	14/64
Clients covered	15/15	15/15
MIP Gap	0.00%	n/a
Solve time (s)	6.85	0.13
Avg profitability	0.6843	0.7547 [†]

[†]SA profitability uses $|\text{cov}(i)|$ (all clients within range) rather than actually assigned clients, so values are not directly comparable to Gurobi.

5.1. Optimal Placement Detail

Loc	Position	Cost	Clients	Profit
0	(0,0)	1.27	1	0.7874
10	(2,1)	3.61	1	0.2770
14	(6,1)	4.04	2	0.4950
19	(3,2)	1.67	0	0.0000
21	(5,2)	4.47	2	0.4474
27	(3,3)	1.34	0	0.0000
28	(4,3)	1.43	0	0.0000
34	(2,4)	1.72	2	1.1628
43	(3,5)	3.32	1	0.3012
49	(1,6)	1.35	4	2.9630
50	(2,6)	5.14	3	0.5837
52	(4,6)	2.30	3	1.3043
53	(5,6)	5.23	3	0.5736

Locations 19, 27, and 28 are placed with zero direct client coverage; they serve as *relay nodes* necessary to maintain

backbone connectivity between geographically separated clusters.

5.2. SA Placement Detail

Loc	Position	Cost	Clients	Profit
0	(0,0)	1.27	1	0.7874
10	(2,1)	3.61	2	0.5540
14	(6,1)	4.04	2	0.4950
19	(3,2)	1.67	0	0.0000
21	(5,2)	4.47	2	0.4474
27	(3,3)	1.34	0	0.0000
28	(4,3)	1.43	0	0.0000
34	(2,4)	1.72	2	1.1628
41	(1,5)	6.51	4	0.6144
43	(3,5)	3.32	1	0.3012
49	(1,6)	1.35	4	2.9630
52	(4,6)	2.30	3	1.3043
54	(6,6)	7.78	3	0.3856
61	(5,7)	1.29	2	1.5504

Node 41 is an additional relay in the SA solution (absent in Gurobi). It carries no client coverage but bridges the upper-left cluster to v_0 , replacing the cheaper nodes 50 and 53 found by Gurobi.

6. Discussion

6.1. Solution Quality

Gurobi certifies an optimal cost of **36.89** (MIP gap 0%). SA finds a feasible solution of cost **42.10**, a **14.1 %** optimality gap, using one extra repeater (14 vs. 13). Both methods achieve full 15/15 client coverage. The shared relay nodes (19, 27, 28) confirm that both approaches independently identify the same mandatory backbone bridges.

6.2. Speed vs. Optimality

SA runs in **0.13 s** versus **6.85 s** for Gurobi — a $\approx 51\times$ speed-up on this small instance. Gurobi’s worst-case complexity is $O(2^{|V|})$; SA scales as $O(T_{\max} \cdot |V| \cdot |J|)$ and is therefore better suited to larger grids where B&B becomes intractable. For $|V| = 64$ and 15 clients, however, the exact solver is already fast enough that SA’s speed advantage is of limited practical value.

6.3. Relay Node Cost

The extra cost in the SA solution (42.10 vs. 36.89) stems primarily from node 41 (cost 6.51), which Gurobi avoids by routing connectivity through the cheaper pair 50+53 (costs 5.14+5.23, shared with client coverage). SA’s greedy repair activates the cheapest *inactive frontier* node at each step, which can miss globally cheaper relay paths.

6.4. SA Limitations & Improvements

The repair-based neighbourhood guarantees feasibility but can over-activate nodes: each bridge insertion is a local greedy choice with no backtracking. Potential improvements include: (i) a 2-opt swap operator (exchange two active nodes) to escape local minima; (ii) adaptive reheating when no improving move is found; (iii) longer cooling schedules or larger T_0 to broaden early exploration. Despite these limitations, SA produces a valid, near-optimal solution at a fraction of the exact solver’s runtime, making it a practical baseline for larger or time-constrained instances.